Reviewer Pack — Minimal Monodromy Group Order and Communication Complexity R...

Entry 82933da8aa2a · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-08 05:46:52 UTC

Minimal Monodromy Group Order and Communication Complexity Rank Variance

Entry ID: `82933da8aa2a` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-08 05:46:52 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Monodromy Theory) **Field B** (complexity object): Communication Complexity (Matrix Rank Variance)

Statement:

> For every Boolean function f: {0,1}^n → {0,1}, the order of its monodromy group |G_f| is linearly correlated with the variance of its communication complexity rank, such that |G_f| = Θ(CommRankVar(f)).

Rationale (proposer's reasoning):

> Monodromy theory studies the symmetries and automorphisms in algebraic varieties, which could potentially reveal hidden structures in Boolean functions that influence their communication complexity. A linear correlation would suggest a direct connection between these properties.

Taxonomy category: `AlgebraicGeometry × CommunicationComplexity` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `d27abc9926c977e5`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> For Boolean functions f: {0,1}^n → {0,1}, |G_f| is considered supported if support_fraction >= 0.8 AND metric_mean ≤ 3, where support_fraction is the proportion of n≤40 functions with |G_f| = Θ(CommRankVar(f)), and metric_mean is the mean value of |G_f| for these functions.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Minimal Monodromy Group Order" AND "Communication Complexity Rank Variance" - "Algebraic Geometry" AND "Matrix Rank Variance" AND "Monodromy Theory" - "Boolean function" AND "monodromy group order" AND "communication complexity"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=124, elapsed=240.4s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity_rank_variance(f):
        n = int(math.log2(len(f)))
        rank = [f.count(i) for i in set(f)]
        mean = sum(rank) / len(rank)
        variance = sum((x - mean)**2 for x in rank) / len(rank)
        return variance

    def monodromy_group_order(f):
        # Placeholder function to simulate computing the monodromy group order
        n = int(math.log2(len(f)))
        return 2**n

    metric_name = "monodromy_group_order"
    instances_tested = 0
    n_max = 0
    total_metric_value = 0.0
    support_count = 0
    counterexample = ""

    for n in [5, 10, 15, 20, 30, 40]:
        if n > n_max:
            n_max = n

        for _ in range(5):
```

```
            f = generate_boolean_function(n)
            instances_tested += 1
            G_f = monodromy_group_order(f)
            CommRankVar_f = communication_complexity_rank_variance(f)

            total_metric_value += G_f

            if abs(G_f - CommRankVar_f) <= 0.1 * max(G_f, CommRankVar_f):
                support_count += 1

    metric_mean = total_metric_value / instances_tested
    support_fraction = support_count / instances_tested

    conjecture_holds = support_fraction >= 0.8 and metric_mean <= 3

    return {
        "metric_name": metric_name,
        "metric_value": metric_mean,
        "instances_tested": instances_tested,
        "n_max": n_max,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    metric_values = [r["metric_value"] for r in results if r["conjecture_holds"]]
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={sum(metric_values)/len(metric_values):.2f} std={math.sqrt(
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={sum(metric_values)/len(metric_values):.2f} std={math.sqrt(
    else:
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_f
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
TIMEOUT after 240s
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

The test timed out before producing data, which means it did not meet the pre-registered support conditions for |G_f| to be considered supported. | next: Re-run the test with increased time limits or optimize the code to ensure it completes within the allowed time frame.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15831
2	preregistration	ollama_remote	glm4:latest	0	0	10764
3	novelty	ollama_remote	glm4:latest	0	0	8655
4	novelty	ollama_remote	glm4:latest	0	0	11404
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	24583
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9755
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8832
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9562
9	judge	ollama_remote	glm4:latest	0	0	39532

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 138919 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/82933da8aa2a.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/82933da8aa2a.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/82933da8aa2a.tar.gz` (if generated)